



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2008

**HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #1**

Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 90 minutes.
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for untidy or badly arranged work.
- Start each **NEW** section in a separate answer booklet.

Total Marks - 90 Marks

- Attempt questions 1 - 3
- All questions are **NOT** of equal value.

Examiner: *E. Choy*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 90

Attempt Questions 1 - 3

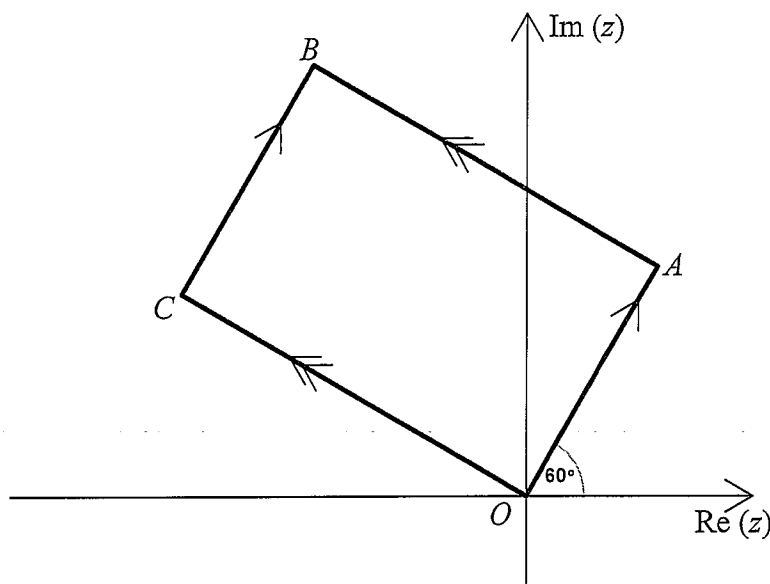
All questions are not of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (30 marks)		Use a SEPARATE writing booklet	Marks
(a)	(i)	For the complex number $z = \sqrt{3} - i$ find	
	(α)	$ z $;	1
	(β)	$\arg z$.	1
	(ii)	For $z = \sqrt{3} - i$, show on an Argand diagram (clearly labelled) $z, \bar{z}, z^2$ and $\frac{1}{z}$.	4
(b)		Sketch on an Argand diagram the region defined by	
	(i)	$ \arg z \leq \frac{\pi}{4}$ and $z + \bar{z} < 6$ and $ z > 3$	3
	(ii)	$\arg\left(\frac{z+2i}{z}\right) = -\frac{\pi}{3}$	2
(c)	(i)	Express $\frac{\sqrt{3}}{2} + \frac{1}{2}i$ in modulus - argument form.	1
	(ii)	Using (i) above, find the value(s) of n such that $\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^n = 1$, where n is a positive integer.	3

Question 1 is continued on page 3

(d)



The above figure shows a parallelogram $OABC$ in an Argand diagram.

$|OA| = 2$ and OA makes an angle of 60° with the positive real axis.

Let z_1 , z_2 and z_3 be the complex numbers represented by vertices A , B and C respectively.

It is given that $z_3 = (\sqrt{3}i)z_1$.

- (i) Find z_1 and z_3 in the form $x + iy$ 4
- (ii) Show that $\frac{z_2}{z_1} = 1 + \sqrt{3}i$. 4
- (iii) Let $\omega = \cos \theta + i \sin \theta$, where $0^\circ \leq \theta < 360^\circ$. Point E is a point on the Argand diagram representing the complex number ωz_3 .
Find the value(s) of θ in each of the following cases:
 - (α) E represents the complex number z_3 ; 3
 - (β) Points E , O and A lie on the same straight line. 4

Question 2 starts on page 4

Question 2 (20 marks)	Use a SEPARATE writing booklet	Marks
(a)	Find the constants p and q such that $x - 2$ is a common factor of $x^3 - x^2 - 2px + 3q$ and $qx^3 - px^2 + x + 2$.	3
(b)	$P(x)$ is a cubic polynomial with real coefficients. One zero of $P(x)$ is $1 + 2i$ and the constant term is -15 . Also, $P(2) = 5$. Write $P(x)$ in the form $ax^3 + bx^2 + cx + d$.	5
(c)	Factorise $4x^4 + 1$ as a product of real quadratic polynomials.	3
(d)	If α and $-\alpha$ are both roots of $x^3 + mx^2 + nx + p = 0$, show that $mn = p$.	4
(e)	(i) Explain briefly why any rational root of $x^n + a_{n-1}x^{n-1} + \dots + a_2x^2 + a_1x + a_0 = 0$, must be an integer, where a_i ($i = 0, 1, 2, \dots, n-1$) are integers.	2
	(ii) Find the integral roots of $x^3 - 6x^2 + 6x + 8 = 0$ and hence find all the roots	3

Question 3 starts on page 5

Question 3 (40 marks)**Use a SEPARATE writing booklet****Marks**

(a) (i) Evaluate $\int_1^5 \frac{dx}{x^2 + 5x + 6}$

3

(ii) Find $\int \frac{1 + \sin x}{1 + \cos x} dx$

3

(iii) Evaluate $\int_{\frac{1}{\sqrt{2}}}^{\frac{\sqrt{3}}{2}} \frac{dx}{x^2 \sqrt{1-x^2}}$

3

(iv) Find $\int x^2 \cos x dx$

3

(b) (i) By means of the substitution $u = a - x$ prove that

2

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

(ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \sqrt{1 - \sin 2x} dx$.

3

(c) Let $y = x^n \sin x$, where n is a positive integer.

(i) Find $\frac{dy}{dx}$

2

(ii) Hence, show that $\int x^n \cos x dx = x^n \sin x - n \int x^{n-1} \sin x dx$, where $n \geq 1$.

2

(iii) Use this result, in (ii) above, to show that $\int_0^\pi x^n \cos x dx = -n \int_0^\pi x^{n-1} \sin x dx$.

2

(iv) Hence evaluate $\int_0^\pi x \cos x dx$.

2

Question 3 continued on page 6

Question 3 continued

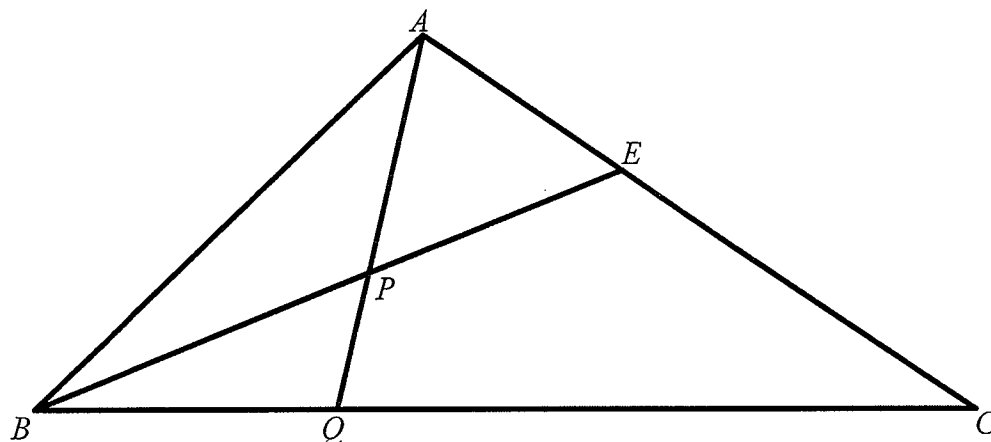
Marks

- (d) How many ways are there to split 4 red, 5 blue and 7 black balls among:
- (i) Two boxes, without any restriction? 2
 - (ii) Two boxes, with no box empty? 2

- (e) In the $\triangle ABC$ below, BE bisects $\angle ABC$. 3

APQ is a straight line such that $AP = AE$.

Prove that AB is a tangent to the circle that passes through the points A , Q and C .



- (f) A straight line is drawn to the curve $y = x^4 - 4x^3 - 18x^2$ so that it is a common tangent at two distinct points on the curve.

- (i) If the equation of the tangent is $y = mx + b$ and its points of contact are $x = p$ and $x = q$, show that

(α) $p + q = 2$; 2

(β) $p^2 q^2 = -b$. 1

- (ii) Hence, or otherwise, find the equation of the common tangent. 5

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

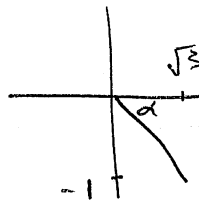
NOTE: $\ln x = \log_e x, x > 0$

Question 1

a) i) $z = \sqrt{3} - i$

$\alpha) |z| = 2$

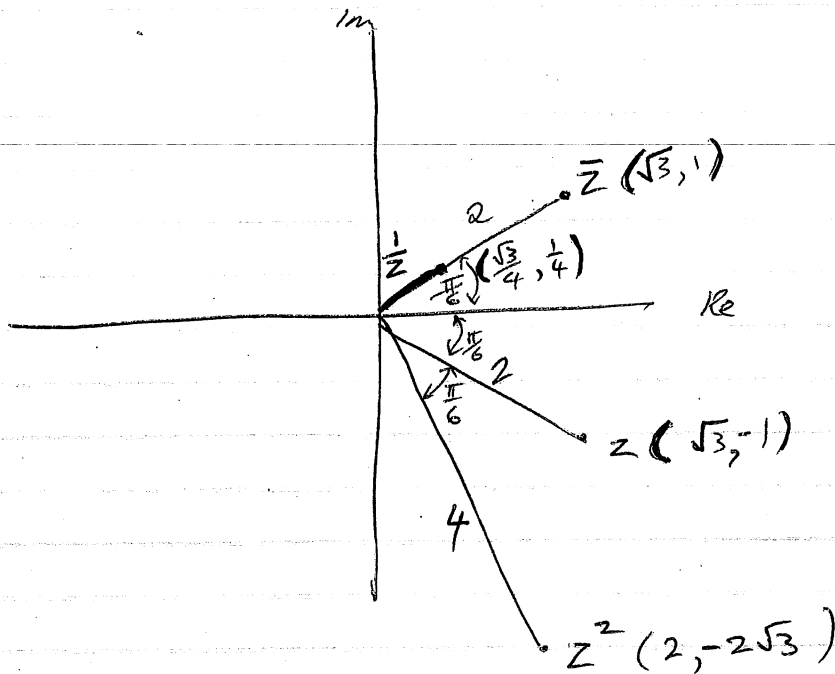
$\beta) \arg z = -\frac{\pi}{6}$



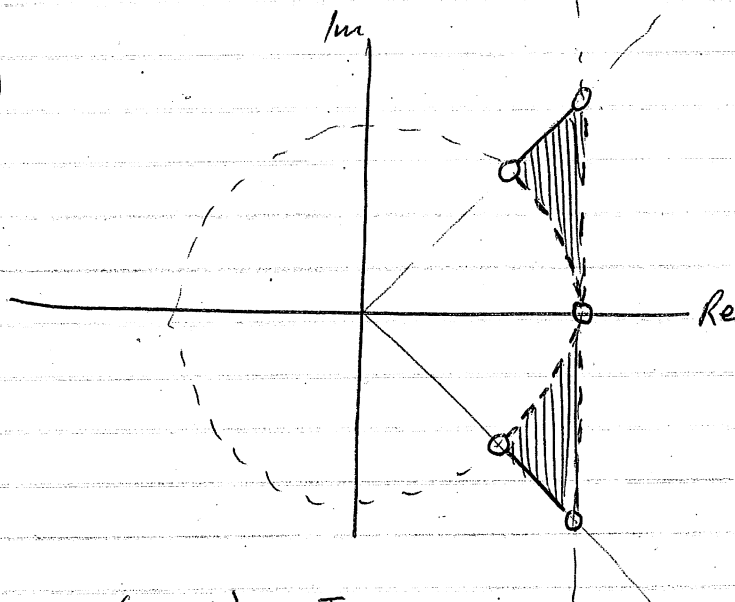
$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

ii)



b) i)



$$|\arg z| \leq \frac{\pi}{4}$$

$$z + \bar{z} < 6$$

$$2x < 6$$

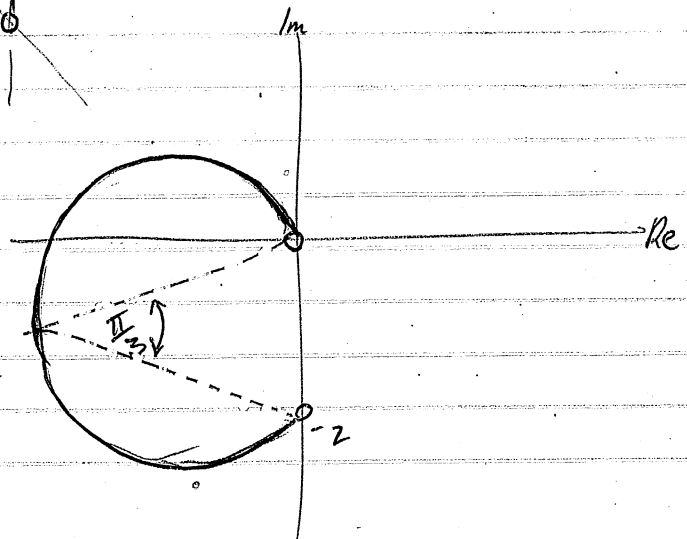
$$x < 3$$

$$|z| > 3$$

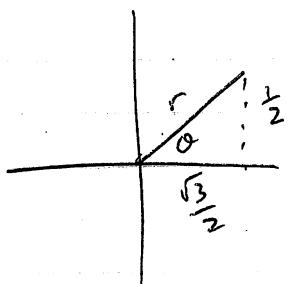
ii) $\arg\left(\frac{z+2i}{z}\right) = -\frac{\pi}{3}$

$$\arg(z+2i) - \arg(z) = -\frac{\pi}{3}$$

$$\arg(z) - \arg(z+2i) = \frac{\pi}{3}$$



c) i)



$$r = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2}$$

$$r = 1$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6}$$

$$\frac{\sqrt{3}}{2} + \frac{1}{2}i = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$$

$$ii) \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^n = 1$$

$$\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^n = \cos(0 + 2k\pi) + i \sin(0 + 2k\pi)$$

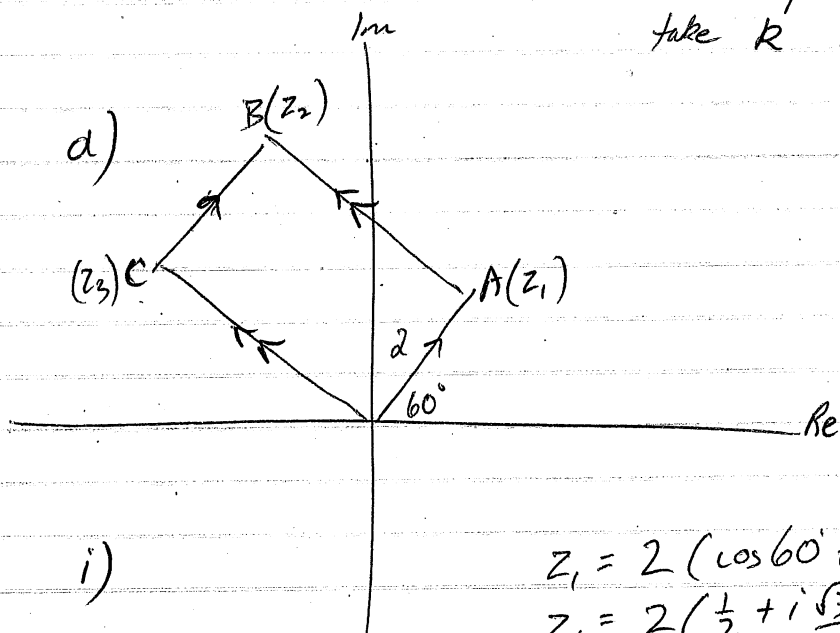
$$\cos \frac{n\pi}{6} + i \sin \frac{n\pi}{6} = \cos(0 + 2k\pi)$$

where k is an integer

$$\frac{n\pi}{6} = 2k\pi$$

$n = 12K$, since n needs to be a positive integer

take K to be a positive integer.



i)

$$z_1 = 2(\cos 60^\circ + i \sin 60^\circ)$$

$$z_1 = 2\left(\frac{1}{2} + i \frac{\sqrt{3}}{2}\right)$$

$$z_1 = 1 + \sqrt{3}i$$

$$\begin{aligned}
 z_3 &= (\sqrt{3}i)z_1 \\
 &= (\sqrt{3}i)(1+\sqrt{3}i) \\
 &= -3+\sqrt{3}i
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } z_2 &= z_1 + z_3 \\
 &= z_1 + (\sqrt{3}i)z_1 \\
 &= z_1(1+\sqrt{3}i)
 \end{aligned}$$

$$\begin{aligned}
 \frac{z_2}{z_1} &= \frac{z_1(1+\sqrt{3}i)}{1+\sqrt{3}i} \\
 &= z_1 \\
 &= 1+\sqrt{3}i
 \end{aligned}$$

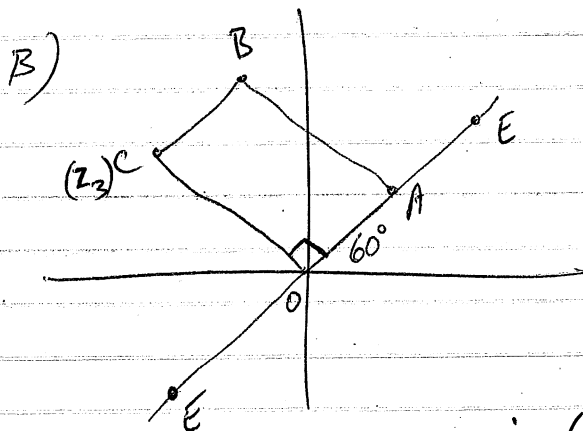
$$\text{iii) a) } \omega = \cos \theta + i \sin \theta \quad 0^\circ \leq \theta < 360^\circ$$

$$\omega z_3 = z_3$$

$$\therefore \omega = 1$$

$$\cos \theta + i \sin \theta = 1$$

$$\theta = 0^\circ$$



since $\arg z_3 = 150^\circ$

$$\angle COA = 90^\circ$$

$$\therefore \theta = 90^\circ, 270^\circ$$

Question 2

(a) $P(x) = x^3 - x^2 - 2px + 3q$
 (3) $P(2) = 2^3 - 2^2 - 4p + 3q = 0$
 $\therefore -4p + 3q = -4$ — (1)

$$Q(x) = qx^3 - px^2 + x + 2$$

$$Q(2) = q(2)^3 - 4p + 4 = 0$$

$$\therefore -4p + 8q = -4$$
 — (2)
 $\Rightarrow q = 0 \text{ and } p = 1$

(b) Since pol real $\Rightarrow 1-2i$ is

(5) also a zero

$\therefore x^2 - 2x + 5$ is quad. factor

$$\Rightarrow P(x) = (kx - 3)(x^2 - 2x + 5)$$

and $P(2) = (2k - 3)(5) = 5$

$$\therefore k = 2$$

$$\text{i.e. } P(x) = (2x - 3)(x^2 - 2x + 5)$$

$$\Rightarrow P(x) = 2x^3 - 7x^2 + 16x - 15$$

(c) $4x^4 + 1 = (4x^4 + 4x^2 + 1) - 4x^2$

$$= (2x^2 + 1)^2 - (2x)^2$$

$$= (2x^2 - 2x + 1)(2x^2 + 2x + 1)$$

(d) ⁴ let $\alpha, -\alpha, \beta$ be roots

$$\text{sum } \alpha + (-\alpha) + \beta = -m \Rightarrow \boxed{\beta = -m}$$

$$\alpha(-\alpha) + \alpha\beta + (-\alpha)\beta = n \Rightarrow \boxed{\alpha^2 = -n}$$

$$\alpha(-\alpha)\beta = -p \Rightarrow \boxed{\alpha^2\beta = p}$$

Subst. $\beta = -m$ in (3)

$$\Rightarrow \alpha^2 = -\frac{p}{m}$$

From (2) $\alpha^2 = -n$

$$\left. \begin{array}{l} \Rightarrow \alpha^2 = -\frac{p}{m} \\ \text{From (2) } \alpha^2 = -n \end{array} \right\} \begin{array}{l} -n = -\frac{p}{m} \\ p = mn \end{array}$$

(e)

(i) If a pol. has a rational root $\frac{p}{q}$ then p divides the constant term and q divides the leading coeff. (2)
 which in this case is 1 $\Rightarrow q = 1$

$\therefore$ root is p

(ii) $P(4) = 0 \Rightarrow (x - 4)$ factor

$$P(x) = (x - 4)(x^2 - 2x - 2)$$

Roots are $x = 4, x = \frac{2 \pm 2\sqrt{3}}{2}$ (3)

i.e. $x = 4, 1 \pm \sqrt{3}$

Q3

(a) (i) Let $\frac{1}{x^2+5x+6} \equiv \frac{A}{x+3} + \frac{B}{x+2}$.

ii. $1 \equiv A(x+2) + B(x+3)$

Let $x = -2$, $\boxed{1 = 13}$

Let $x = -3$, $1 = -A \Rightarrow \boxed{A = -1}$

Hence,
$$\begin{aligned} \int_1^5 \frac{dx}{x^2+5x+6} &= \int_1^5 \left(\frac{1}{x+2} - \frac{1}{x+3} \right) dx \\ &= \left[\ln(x+2) - \ln(x+3) \right]_1^5 \\ &= \ln 7 - \ln 8 - (\ln 3 - \ln 4) \\ &= \ln \frac{7 \times 4}{3 \times 8} \\ &= \boxed{\ln \frac{7}{6}}. \end{aligned}$$

(ii)
$$\begin{aligned} \int \frac{1 + \sin x}{1 + \cos x} dx &= \int \frac{1 + \frac{2t}{1+t^2}}{1 - \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2} \quad \text{where } t = \tan \frac{x}{2} \\ &= \int \frac{1+t^2+2t}{1+t^2-(1-t^2)} \times \frac{2dt}{1+t^2} \\ &= \int \frac{1+t^2+2t}{2(1+t^2)} \cdot 2dt \\ &= \int \left(\frac{1+t^2}{1+t^2} + \frac{2t}{1+t^2} \right) dt \\ &= \int \left(1 + \frac{2t}{1+t^2} \right) dt \\ &= t + \ln(1+t^2) + C \\ &= \boxed{\tan \frac{x}{2} + \ln(1 + \tan^2 \frac{x}{2}) + C}. \end{aligned}$$

(iii)

$$\int_{\frac{1}{\sqrt{2}}}^{\frac{\sqrt{3}}{2}} \frac{dx}{x^2 \sqrt{1-x^2}}$$

$$\text{let } x = \cos \theta$$

$$dx = -\sin \theta d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{6}} \frac{-\sin \theta d\theta}{\cos^2 \theta \sqrt{1-\cos^2 \theta}}$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sin \theta d\theta}{\cos^2 \theta \cdot \sin \theta}$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{d\theta}{\cos^2 \theta}$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sec^2 \theta d\theta$$

$$= \left[\tan \theta \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$$

$$= 1 - \frac{1}{\sqrt{3}}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}} \quad \text{or} \quad \left(\frac{3-\sqrt{3}}{3} \right)$$

$$(iv) \quad I = \int x^2 \cos x dx$$

$$= \int x^2 \frac{d}{dx} (\sin x) dx$$

$$= x^2 \sin x - \int 2x \sin x dx$$

$$= x^2 \sin x - 2 \int x \frac{d}{dx} (-\cos x) dx$$

$$= x^2 \sin x - 2 \left[-x \cos x - \int -\cos x dx \right]$$

$$= x^2 \sin x - 2 \left[-x \cos x + \int \cos x dx \right]$$

$$= x^2 \sin x - 2 \left[-x \cos x + \sin x \right]$$

$$= x^2 \sin x + 2x \cos x - 2 \sin x + C$$

(b) (i)

$$\text{R.T.P } \int_0^a f(x) dx = \int_0^a f(a-x) dx.$$

$$\begin{aligned} \underline{\text{LHS}} &= \int_0^a f(x) dx && \text{let } u = a-x. \\ & && \therefore x = a-u. \\ & && dx = -du \\ &= - \int_a^0 f(a-u) du. \\ &= \int_0^a f(a-x) dx. \\ &= \underline{\text{RHS}}. \end{aligned}$$

$$\begin{aligned} \text{(ii)} \int_0^{\frac{\pi}{4}} \sqrt{1 - \sin 2x} dx &= \int_0^{\frac{\pi}{4}} \sqrt{1 - \sin 2(\frac{\pi}{4} - x)} dx \\ &= \int_0^{\frac{\pi}{4}} \sqrt{1 - \sin(\frac{\pi}{2} - u)} du \\ &= \int_0^{\frac{\pi}{4}} \sqrt{1 - \cos 2x} dx \\ &= \int_0^{\frac{\pi}{4}} \sqrt{1 - (1 - 2\sin^2 x)} dx \\ &= \int_0^{\frac{\pi}{4}} \sqrt{2\sin^2 x} dx. \\ &= \sqrt{2} \int_0^{\frac{\pi}{4}} \sin x dx. \\ &= \sqrt{2} \left[-\cos x \right]_0^{\frac{\pi}{4}} \\ &= \sqrt{2} \left[-\frac{1}{\sqrt{2}} - -1 \right] \\ &= -1 + \sqrt{2} \\ &= \boxed{\sqrt{2} - 1} \end{aligned}$$

(c) (i) $\frac{d}{dx}(x^n \sin x) = x^n \cos x + n x^{n-1} \sin x$

(ii) now $\int (x^n \cos x + n x^{n-1} \sin x) dx = x^n \sin x$

$\therefore \int x^n \cos x dx + n \int x^{n-1} \sin x dx = x^n \sin x$

$\therefore \int x^n \cos x dx = x^n \sin x - n \int x^{n-1} \sin x dx$

(iii) $\therefore \int_0^\pi x^n \cos x dx = \left[x^n \sin x \right]_0^\pi - n \int_0^\pi x^{n-1} \sin x dx$
 $= 0 - 0 - n \int_0^\pi x^{n-1} \sin x dx$

$\therefore \int_0^\pi x^n \cos x dx = -n \int_0^\pi x^{n-1} \sin x dx$

(iv). let $n = 1$

$\int_0^\pi x \cos x dx = - \int_0^\pi x^0 \sin x dx$

$= - \left[-\cos x \right]_0^\pi$

$= \left[\cos x \right]_0^\pi$

$= \cos \pi - \cos 0$

$= -1 - 1$

$= \boxed{-2}$

(d)

$$(i) \binom{5}{1} \times \binom{6}{1} \times \binom{8}{1} = 5 \times 6 \times 8 = \boxed{240}$$

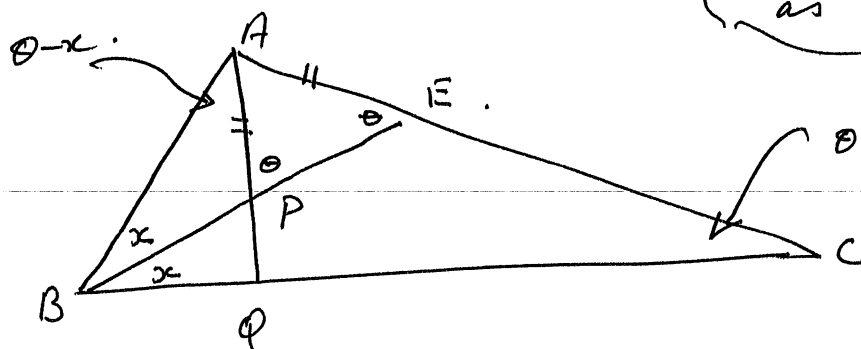
$$(ii) 240 - 2 = \boxed{238}$$

NB if the number of balls were a, b & c , then answer is

$$\binom{a+1}{1} \times \binom{b+1}{1} \times \binom{c+1}{1}$$

which is not the same as $a+b+c$. so ${}^1p_r = 240$ was unacceptable

(e)



Since $\triangle APE$ is isosceles

$$\angle APE = \angle AEP = \theta \quad (\text{base angles of an isosceles triangle are equal})$$

$$\angle ABE = \angle ACE \quad (\text{data})$$

$$\text{Now } \angle BAQ = \theta - x \quad (\text{exterior angle equal to the sum of the interior opposite angles})$$

$$\text{Similarly } \angle BCA = \theta - x \quad (" ")$$

$$\therefore \angle BAQ = \angle BCA$$

ie angle between tangent and chord is equal to the angle in the alternate segment
[NB actually the converse is required - didn't expect a proof!]

(f) (i) Let $y = x^4 - 4x^3 - 18x^2 = mx + b$.

$$\Rightarrow x^4 - 4x^3 - 18x^2 - mx - b = 0$$

has a common tangent at two distinct points. (then these roots will be double roots.)

ie. p, p, q, q .

now $S_1 = p + p + q + q = 4$ (ie. $-\frac{b}{a}$)

$$(2) \quad \begin{array}{l} 2(p+q) = 4 \\ \boxed{p+q = 2} \end{array} \quad (A)$$

$$S_4 = p \times p \times q \times q = -b$$

$$(3) \quad \text{ie } \boxed{p^2 q^2 = -b} \quad (B) \quad \left(\text{ie } \frac{c}{a}\right)$$

(ii) now $S_2 = \boxed{p^2 + q^2 + 4pq = -18} \quad (C) \quad \left(\text{ie } \frac{c}{a}\right)$

$$S_3 = 2p^2 q + 2q^2 p = m \quad \left(\text{ie } -\frac{d}{a}\right)$$

$$2pq(p+q) = m$$

$$4pq = m$$

$$\boxed{pq = \frac{m}{4}}$$

From (C) $(p+q)^2 + 2pq = -18$.

$$4 + \frac{m}{2} = -18$$

$$\frac{m}{2} = -22$$

$$\boxed{m = -44}$$

$$\therefore pq = -\frac{44}{4} = -11 \quad \therefore (-11)^2 = -b$$

$$b = -121$$

$$\therefore \boxed{y = -44x - 121}$$